A Technical Introduction to Artificial neural network

$$a^{(l)} = \sigma!\,(W^{(l)}a^{(l-1)} + b^{(l)})$$

Section 1

Supervised learning Supervised learning pairs inputs and desired outputs. The learning task is to produce the desired output for each input. In this case, the cost is related to eliminating incorrect outputs. A commonly used cost is the mean-squared error, which tries to minimize the average squared error between the network's output and the desired output. Tasks suited for supervised learning include pattern recognition (classification) and regression (function approximation). Supervised learning is applicable to sequential data (e.g., for handwriting, speech and gesture recognition). This can be thought of as learning with a "teacher", in the form of a function that provides continuous feedback.

Section 2

NNs are composed of digital neurons, conceptually derived from biological neurons. Each neuron has one or more numerical inputs and produces a single numerical output. An input (e.g., an image) is typically parceled across the set of input neurons (each getting a piece of the image). Each neuron's inputs, weighted by the weights of the connections from each of those inputs are summed. A bias term is added to this sum. bias) The result is then passed through a nonlinear activation function to produce the neuron's output. Small enough outputs may be zeroed out (ignored).

Section 3

Modern deep learning Between 2009 and 2012, NNs began winning prizes in image recognition contests, approaching human performance on various tasks, initially in pattern recognition and handwriting recognition. In 2011, DanNet, a CNN by Dan Ciresan, Ueli Meier, Jonathan Masci, Luca Maria Gambardella, and Schmidhuber achieved for the first time superhuman performance in a visual pattern recognition contest, outperforming traditional methods by a factor of 3. It then won more contests. They also showed how max-pooling CNNs on GPU improved performance. In October 2012, AlexNet by Alex Krizhevsky, Ilya Sutskever, and Hinton won the large-scale ImageNet competition by a significant margin over shallow machine learning methods. Further incremental improvements included the VGG-16 network by Karen Simonyan and Andrew Zisserman and Google's Inceptionv3. In 2012, Ng and Dean created a network that learned to recognize higher-level concepts, such as cats, from training only on unlabeled images. Unsupervised pre-training and increased computing power from GPUs and distributed computing allowed the use of larger networks, particularly in image and visual recognition problems, which became known as "deep learning". Radial basis function and wavelet networks were introduced in 2013. These can be shown to offer best approximation properties and have been applied in nonlinear system identification and classification applications. Generative adversarial networks (GANs) (Ian Goodfellow et al., 2014) became state of the art in generative modeling from 2014–2018. GAN was originally published in 1991 by Schmidhuber, who called it "artificial curiosity": two neural networks contest with each other in the form of a zero-sum game, where one network's gain is the other network's loss. The first network is a generative model that models a probability distribution over output patterns. The second network learns by gradient descent to predict the reactions of the environment to these patterns. Excellent image quality was achieved by Nvidia's StyleGAN (2018) based on the Progressive GAN by Tero Karras et al. Here, the GAN generator is grown from small to large scale in a pyramidal fashion. Image generation by GAN reached popular success, and provoked discussions concerning deepfakes. Diffusion models (2015) eclipsed GANs in generative modeling thereafter, with systems such as DALL·E 2 (2022) and Stable Diffusion (2022). In 2014, the state of the art was training "[a] very deep neural network" with 20 to 30 layers. Stacking too many layers led to a steep reduction in training accuracy, known as the "degradation" problem. In 2015, training very deep networks advanced with the highway network, published in May, and the residual neural network (ResNet) in December. ResNet behaves like an open-gated Highway Net.

Section 4

Nodes are arranged in layers, with the bottom layers directly addressing raw data, while the top layers reveal the final results. Intermediate layers gradually increase the level of abstraction, so what began as for example, pixels in an image, gradually resolves into things such as object boundaries, and then into real-world objects such as letters and faces. Single layer and unlayered networks are also used. Multiple connection patterns have been used. Traditionally, they are fully connected, with every neuron in one layer connecting to every neuron in the next layer. However, in convolutional neural networks, some layers are convolutional, meaning each neuron in one layer is connected to a subset of neurons in the previous layer, such as those representing one section of an image. In most neural networks, the outputs of the neurons in one layer are connected only to neurons in the immediately following layer (directed acyclic graph), meaning information only flows forward from one layer to the next.

These are known as feedforward networks. In contrast, networks that allow connections between neurons in the same or previous layers are known as recurrent networks.

Section 5

Other In a Bayesian framework, a distribution over the set of allowed models is chosen to minimize cost. Evolutionary methods, gene expression programming, simulated annealing, expectation–maximization, non-parametric methods and particle swarm optimization are other learning algorithms. Convergent recursion is a learning algorithm for cerebellar model articulation controller (CMAC) neural networks.

Section 6

Medicine NNs analyze medical datasets. They enhance diagnostic accuracy, especially by interpreting complex medical imaging for early disease detection, and by predicting patient outcomes for personalized treatment planning. In drug discovery, NNs speed up the identification of potential drug candidates and predict their efficacy and safety, significantly reducing development time and costs. Additionally, their application in personalized medicine and healthcare data analysis allows tailored therapies and efficient patient care management.

Section 7

Lack of interpretability NNs are "black box" systems that have complicated understanding of their decision-making processes. NNs are further vulnerable to adversarial examples ("poison"), that can cause incorrect predictions. These concerns have led to increased research in explainable artificial intelligence (XAI), robust machine learning, and hybrid AI approaches that combine neural learning with symbolic reasoning. Advocates of hybrid models also say that such a mixture can better capture the mechanisms of the human mind. Progress has included local vs. non-local learning and shallow vs. deep architectures. Analyzing human thought (a biological neural network) has also proven difficult.